25PY101: Engineering Physics

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Summative preparation

Opto-electronics, QM

Instructions

- Answer all questions.
- Show all the calculation steps clearly.
- Draw neat and properly labeled diagrams.
- Use SI units wherever necessary.

Questions

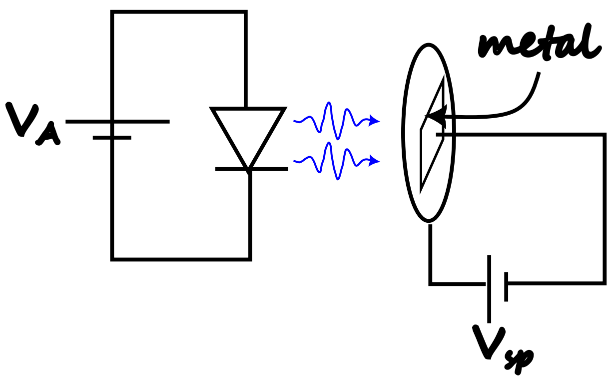
Q1. Electroluminescence ⇒ Photo-electric effect

A blue LED acts as a source of photons for a photoelectric effect experiment. The blue photons shine on the metal to eject photo-electrons. The stopping potential is measured by setting the photo-current to zero.

(a) What is minimum voltage of battery V_A required to carry out the experiment?

(b) If the photocurrent is set to zero at a stopping potential of $V_{sp} = 0.5\,\text{V}$, what is the work function of the metal?

(c) If the work function of the metal is $\phi = 3\,\text{eV}$, what is the photocurrent?



Q2. Graphical / Sketch-Based Question

Sketch and label the following

1. circuit diagram of LED under forward bias.
2. I-V characteristics of solar cell.

Draw the circuit diagram and label the quantities.

Answer

Aim:

Topic:

Given Quantities:

Working Formulas:

Steps: